Final Exam Solutions
ELEC50014 - Software Systems

Time: 2 Hours
Total Marks: 60

Question 1

(a) [10 marks]

Solved on next page

(b) [5 marks]

Researcher(researcher_id, email, fname, lname)
Affiliations(researcher_id, institution_name)
Track(title, conference_id)
Sub-track(title, parent_track_title)
Conference(conference_id, name)
Submission(researcher_id, paper_id, conference_id, date)
Paper(paper_id, title, abstract)
Authors(researcher_id, paper_id)
Review(review_id, rating, paper_id)
Comments(review_id, comment_text)
ConductsReview(researcher_id, review_id)
Revision(paper_id, number, date, file_link)

(c) [10 marks (5 + 5)]

(i)

```
SELECT COUNT(Appointments.appointmentID)
FROM
Doctors JOIN Appointments ON Doctors.doctorID = Appointments.doctorID
WHERE Doctors.name = doctor_name
AND Appointments.appointmentDate >= db1
AND Appointments.appointmentDate <= db2;
```

(ii)

```
SELECT name FROM Patients WHERE patientID IN (
    SELECT DISTINCT patientID FROM Appointments
    WHERE doctorID IN (
        SELECT doctorID FROM Doctors WHERE speciality = 'Cardiology'
    )
);
```

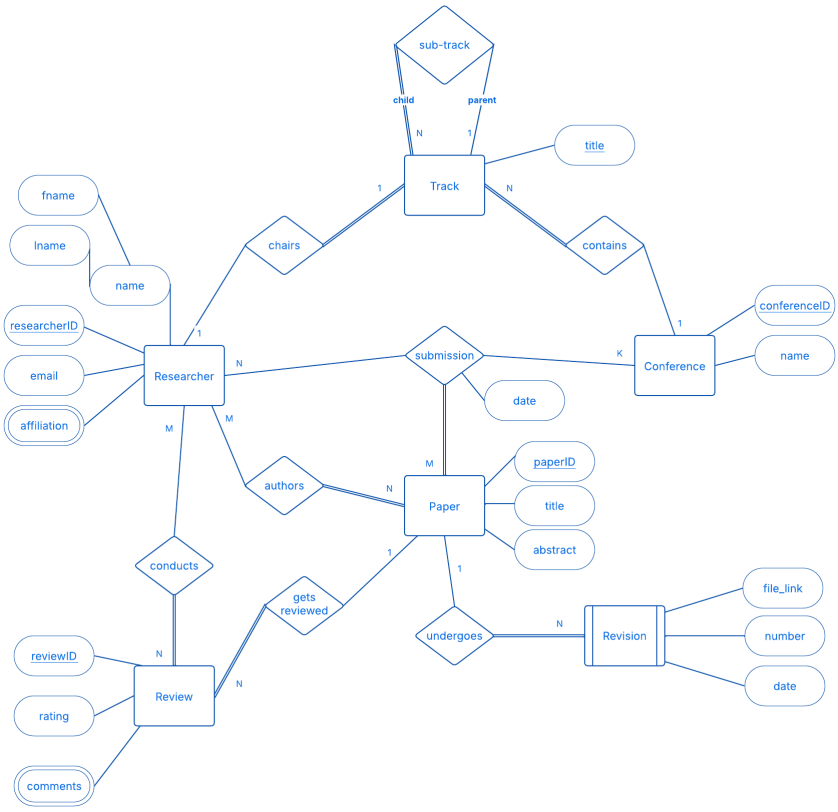


Figure 1: ER Diagram Question 1(a)

Question 2

(a) [6 marks (3+3)]

(i) Iterative DNS resolution is preferred because it reduces the processing burden on DNS servers and improves scalability. In iterative resolution, the client resolver queries each DNS server step by step, rather than relying on a single server to fetch the full response. This reduces the load on upstream DNS servers. Furthermore, caching overhead on upstream resolvers is mitigated, as intermediate results can be stored at the client resolver and reused for future queries, reducing redundant lookups and improving response times.

(ii) HTTP is stateless because each request from the client to the server is independent on the application layer, meaning the server does not retain information about previous requests at this level. While TCP provides reliable, connection-oriented communication, it only ensures correct data transfer, not state retention at the application layer. Since HTTP does not inherently remember past interactions, additional mechanisms such as cookies, sessions, or tokens are required to maintain user state across multiple requests.

(b) [8 marks]

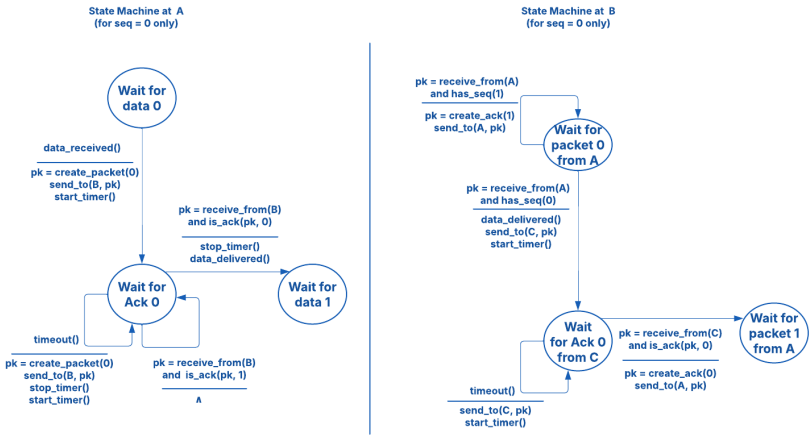


Figure 2: State Diagrams Question 2(b)

(c) [6 marks]

The average end-to-end delay across the three packets is given by:

$$t_{\text{avg}} = \frac{l}{r_A} + \frac{d_1}{v} + 2\frac{l}{r_R} + \frac{d_2}{v}$$

where:

l is the packet size (in bits), r_A is the transmission rate on A's outgoing link (bps), r_R is the transmission rate on R's outgoing link (bps), d_1 is the length of the A–R link (meters), d_2 is the length of the R–B link (meters), v is the propagation speed.

Question 3

(a) [5 marks]

The given network prefix 192.168.16.0/21 provides 2048 IP addresses, ranging from 192.168.16.0 to 192.168.23.255. To allocate subnets, Subnet1 requires 1010 interfaces, while Subnet2 and Subnet3 each require 400 interfaces. The closest power-of-two subnet that accommodates Subnet1 is a /22, which provides 1024 addresses. The nearest suitable subnet for Subnet2 and Subnet3 is a /23, each offering 512 addresses. To satisfy these constraints within the given range, Subnet1 is assigned 192.168.16.0/22, covering

addresses from 192.168.16.0 to 192.168.19.255. Subnet2 is assigned 192.168.20.0/23, covering 192.168.20.0 to 192.168.21.255, and Subnet3 is assigned 192.168.22.0/23, covering 192.168.22.0 to 192.168.23.255. This allocation ensures that all three subnets fit within the given /21 prefix while meeting the required interface constraints.

(b) [4 marks] The time complexity of computing the routing table for router R is:

- Link-state routing (Dijkstra's algorithm): $O(nk)$
- Distance-vector routing (Bellman-Ford algorithm): $O(nk)$

In both cases, R determines the shortest path to each of the n routers by comparing distances from its k neighbors. This requires $O(k)$ operations per destination, leading to a total complexity of $O(nk)$.

(c) [6 marks (3+3)]

(i) ARP uses broadcasts to discover the MAC address associated with a known IP address within a local network, as it doesn't know which device owns the target IP address. Similarly, DHCP clients broadcast requests to locate available DHCP servers because they initially lack an IP address and do not know the server's location. This broadcasting ensures that the necessary network devices receive the requests and can respond appropriately.

(ii) A NAT-enabled router uses an insert function to map a private IP and port to a unique public port for a given destination IP and port. The lookup function retrieves the private IP and port using only the public port when a response arrives. This allows multiple internal hosts to share a single public IP while ensuring correct packet forwarding.